

Core API Endpoints

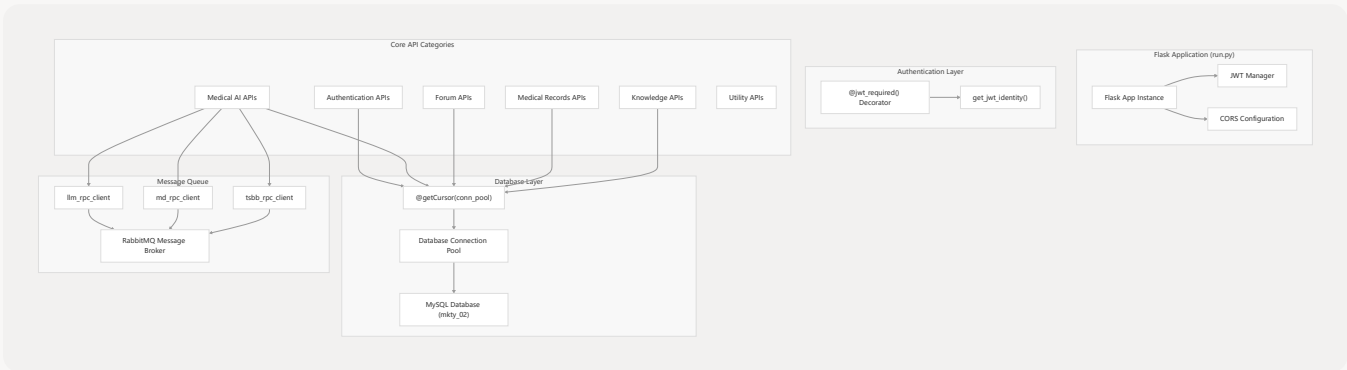
> Relevant source files

This document covers the main Flask-based REST API endpoints that form the backbone of the MKTY Smart Healthcare System backend services. These endpoints handle user management, medical consultations, forum interactions, knowledge management, and AI service integration. For AI model inference services and message queue consumers, see [AI & Machine Learning Services](#). For administrative APIs, see [Admin Backend API](#).

API Architecture Overview

The core API is implemented as a Flask web application that serves as the primary interface between the frontend and all backend services. The API follows a standardized JSON request/response format with JWT-based authentication.

Flask Application Structure



Sources: `backend/run.py` 91–100 `backend/run.py` 83–88

Authentication System

The API uses JWT (JSON Web Tokens) for stateless authentication. Most endpoints require valid authentication tokens in the `Authorization` header.

Authentication Endpoints

Endpoint	Method	Purpose	Authentication Required
<code>/api/loginVerification</code>	POST	User login and token generation	No
<code>/api/register</code>	POST	New user registration	No

Endpoint	Method	Purpose	Authentication Required
/api/test	POST	Token validation test	Yes

JWT Configuration

The JWT system is configured with a strong password for token encryption and tokens are set to never expire for user convenience.

Authentication Flow

loginVerification()



create_access_token()



Frontend Token Storage



Authorization: Bearer



@jwt_required() Decorator



get_jwt_identity()

Sources: `backend/run.py` 93–95 `backend/run.py` 118–158 `backend/run.py` 159–225

User Management APIs

These endpoints handle user profiles, avatars, and account settings.

Endpoint	Method	Purpose	Key Parameters
<code>/api/getUserAvatar</code>	POST	Retrieve user avatar	<code>userId</code>
<code>/api/getUserInfo</code>	POST	Get user profile information	<code>userId</code>
<code>/api/modifyUserInfo</code>	POST	Update user profile	User profile fields
<code>/api/modifyUserAvatar</code>	POST	Update user avatar	<code>userAvatar</code> (base64)
<code>/api/modifyUserPassword</code>	POST	Change user password	<code>userOldPassword</code> , <code>userNewPassword</code>

Sources: `backend/run.py` 226–253 `backend/run.py` 254–293 `backend/run.py` 294–357

Medical AI Service APIs

These endpoints interface with the AI inference services through message queue systems for medical diagnosis and consultation.

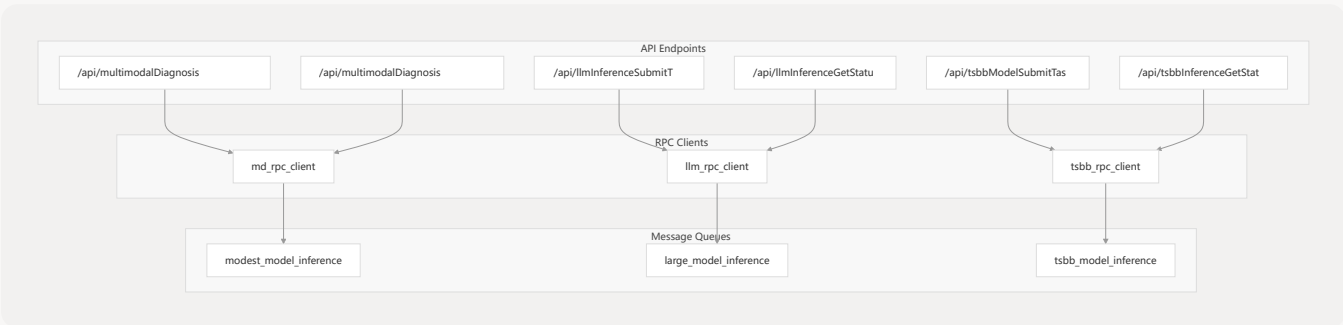
AI Task Submission Endpoints

Endpoint	Method	Purpose	Queue Target
/api/multimodalDiagnosisSubmitTask	POST	Submit multimodal diagnosis task	modest_model_inference
/api/llmInferenceSubmitTask	POST	Submit LLM inference task	large_model_inference
/api/tsbbModelSubmitTask	POST	Submit time series/BigBird task	tsbb_model_inference

AI Task Status Endpoints

Endpoint	Method	Purpose	Returns
/api/multimodalDiagnosisGetStatus	POST	Get multimodal task status	Task status and results
/api/llmInferenceGetStatus	POST	Get LLM task status	Task status and results
/api/tsbbInferenceGetStatus	POST	Get TSBB task status	Task status and results

AI Service Integration Architecture



Sources: `backend/run.py` 512-547 `backend/run.py` 714-741 `backend/run.py` 1394-1439

Medical Records Management APIs

These endpoints manage medical records between doctors and patients with proper access control.

Endpoint	Method	Purpose	Access Control
/api/getMedicalRecords	POST	Get user's medical records list	Patient sees own, Doctor sees assigned
/api/getMedicalRecord	POST	Get specific medical record	Patient/Doctor access verification

Endpoint	Method	Purpose	Access Control
/api/createMedicalRecord	POST	Create new medical record	Doctors only
/api/updateMedicalRecord	POST	Update existing record	Assigned doctor only

Sources: `backend/run.py` 1837-1898 `backend/run.py` 1900-1959
`backend/run.py` 1961-2019

Forum and Community APIs

These endpoints manage the forum system for medical discussions and knowledge sharing.

Forum Management

Endpoint	Method	Purpose	Parameters
/api/addForum	POST	Create new forum	forumName , forumType , forumPermission
/api/getForumList	POST	List forums with filters	forumType , forumPermission
/api/getForumInfo	POST	Get forum metadata	forumId
/api/modifyForumType	POST	Change forum type	forumId , forumType
/api/deleteForum	POST	Delete forum	forumId

Post Management

Endpoint	Method	Purpose	Parameters
/api/sendPost	POST	Create forum post	forumId , postContent , postImagesBase64List
/api/getPostList	POST	Get post IDs for forum	forumId
/api/getPostContent	POST	Get specific post content	postId
/api/praisePost	POST	Like a post	postId
/api/deletePost	POST	Delete own post	postId

Sources: `backend/run.py` 903-946 `backend/run.py` 1112-1166 `backend/run.py` 1168-1193

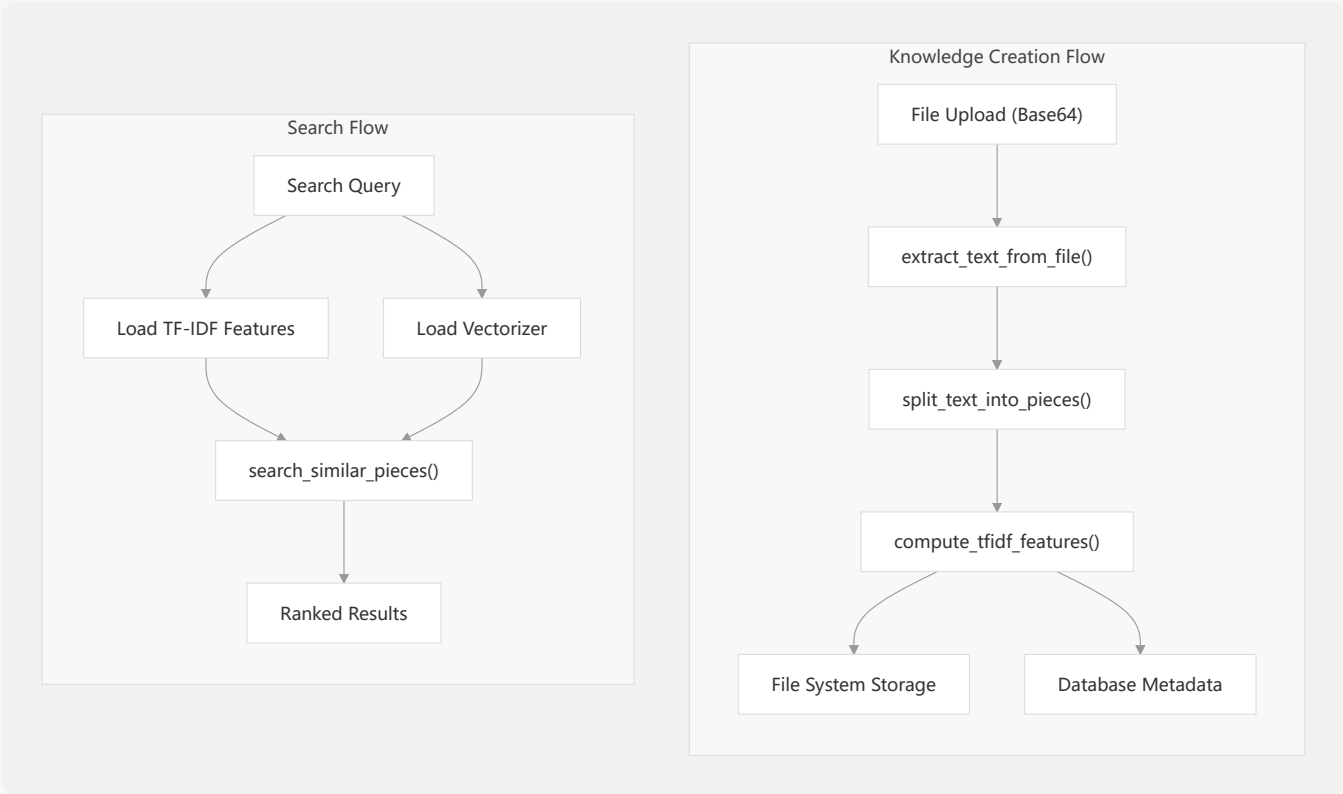
Knowledge Management APIs

These endpoints handle the knowledge base system with document upload, search, and retrieval capabilities.

Endpoint	Method	Purpose	Parameters
/api/knowledge/create	POST	Create knowledge entity	keName , keAbstract , fileContent , fileType
/api/knowledge/search	POST	Search knowledge entities	keyword
/api/knowledge/download/<int:ke_id>	GET/POST	Download knowledge file	ke_id (URL parameter)
/api/knowledge/favorite	POST	Favorite knowledge entity	keId
/api/knowledge/favorites	GET	Get user favorites	None
/api/knowledge/search_pieces	POST	Search within knowledge entity	keId , queryText , topK

Knowledge Entity Processing

The knowledge management system includes sophisticated text processing capabilities:



Task Management APIs

These endpoints manage medical task lists and important items for users.

Endpoint	Method	Purpose	Parameters
/api/addImportantItem	POST	Add medical task	listItemContent , listItemTimeMode , scheduling params
/api/getImportantList	POST	Get user's task list	None (user from JWT)
/api/deleteImportantItem	POST	Delete task	listItemId
/api/finishImportantItem	POST	Mark task complete/incomplete	listItemId , listItemIsFinished

Sources: backend/run.py 575-624 backend/run.py 651-668 backend/run.py 670-697

LLM Session Management APIs

These endpoints manage conversational sessions with the MKTY-3B-Chat language model.